USB PD CTS ENGINEERING CHANGE NOTICE

NOTICE: Any Company or Companies submitting a USB Power Delivery ECR proposal must be one of the following: a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum. If a group of Companies is submitting an ECR proposal, each company must be either a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum.

SPECIFICATION REVISIONS AND ADDENDA: At any point in time, there shall only be one current version of the USB PD CTS, termed the production version. At the same time, there may also be proposed revisions to the specification's design which are not yet approved and shall be held confidential as deemed necessary by the USB 3.0 and USB 2.0 Promoters and within the Group of Working Committee(s).

PROCEDURES FOR SUBMITTING PROPOSALS: Both members of the USB Implementers Forum as a whole and members of the USB 3.0 and USB 2.0 Promoters may submit requests to revise the USB PD CTS Specification. Such a request may be rejected or may result in a USB PD Engineering Change Notice (ECN), which is the official way USB specifications may be changed.

FORMAT OF PROPOSAL: The originator of a request to alter the USB PD CTS Specification may do so by posting this to the USB Power Delivery Compliance working group for review. Once the proposal has been reviewed by the working group it will be passed to the USB 3.0 and 2.0 Promoters for approval to publish.

RESUBMISSION AND APPEAL: The originator of a request that was not approved can redraft the original request. Rewritten proposal will be treated as a new proposal and will be evaluated using the procedures described above. The originator of a request that was not approved can also submit an appeal to the USB 3.0 and 2.0 Promoters. The appeal must be made in writing and addressed to the Secretary of the USB Implementers Forum.

ABOUT THE ENGINEERING CHANGE REQUEST FORM:
The Purpose of this Engineering Change Request Form is to expedite the review process of the proposal by providing explanations, background information, and examples of the proposed changes at a high level. This form serves as an executive summary to the actual proposal.

STEPS ON HOW TO SUBMIT A USB PD ENGINEERING CHANGE REQUEST:

1) Please fill out the Engineering Change Request Form on the following pages completely:
   a) Detail the names and contact details for each of the ECR contributors
   b) Update the ECR Title
   c) Give a minimum of 2-3 sentences for each description on the form outlining the background to the ECR
2) For each section/table/figure to be updated:
   a) Detail the section number, starting page and figure/table number to be updated as appropriate.
   b) Detail existing text under "From Text"
   c) Detail changed text under "To Text"
3) Save the file as "USB PD CTS 1.0 R 1" followed by the ECR Title as per step 1)b)
4) Post the ECR in the USB PD CTS Documents section under "ECR | New ECRs".
   a) This ECR will then be reviewed by the Power Delivery Compliance Working Group.
   b) Revisions to the ECR originating from the review should be submitted as document revision of the original ECR using "Add new document".

USB PD CTS ENGINEERING CHANGE NOTICE

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USB PD CTS ENGINEERING CHANGE NOTICE

Title: Warning in the EPR section.

Applied to: USB PD CTS Specification Q3 2026

Brief description of the functional changes proposed:
Remove warning from the EPR test to be able to test fully the EPR section.

Benefits as a result of the proposed changes:

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:

An analysis of the hardware implications:

An analysis of the software implications:

An analysis of the compliance testing implications:

An analysis of the Vendor Info File (VIF) implications:

USB PD CTS ENGINEERING CHANGE NOTICE

Actual Change Requested

(a). TEST.PD.PS.EPR.SRC3.1 Multiple EPR Request Load Test page 260

From Text:

Description: As a Sink only, the Tester verifies that the Source UUT responds correctly to various *EPR_Request* Messages.
Test Specific Tester Behavior:
Test Conditions:

This test is applicable if the VIF parameter EPR_Supported_*As_Src* is set to Yes.

	Consumer Only	Provider Only, P/C, C/P, DRP
Rev3ChkdSrc		✓

Test Requirements: If Captive_Cable = YES, *vSrcNew* min or *vAvsNew* min limit shall be (*vSrcNew* min) - (0.75*I/X) or (vAvsNew min) - (0.75*I/X), where I is the actual current being drawn, and X=3 if the sourced PDO contract is within the range (0, 3A], or X=5 if the sourced PDO contract is within the range (3A, 5A)

Test Procedures:

1. There are 2 possible bring-up procedures:
    a. The UUT has VIF field PD_Port_Type set to Consumer/Provider, the Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.7.
    b. The UUT has VIF field PD_Port_Type set to anything else, the Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.1.
2. The check fails if the "EPR mode capable" field is set to 0 in the *Source_Capabilities* Message sent by UUT during the bring up procedure and the test stops here. [TEST.PD.PS.EPR.SRC3.1#1]
3. The Tester sends an *EPR_Mode* (Enter) Message with the Action field set to 0x01(Enter) and Data field set to 140 (140W) and enters EPR mode using common procedure COMMON.PROC.PD3.5. [TEST.PD.PS.EPR.SRC3.1#2]
4. The Tester repeats the following steps for each of the EPR_PDOs only (in the EPR_*Source_Capabilities* Message), starting with the 8th PDO:
    a. The Tester sends an *EPR_Request* Message.
        i. B27 (Reserved) = 0b
        ii. B26 (Capability Mismatch) = 0b
        iii. B25 (USB Communication Capable) = 0b
        iv. B24 (No USB Suspend) = 1b
        v. B22 (EPR Mode Capable) = 1b
        vi. For Fixed PDO:
            1. B19…10 (Operating Current):
                a. For first *EPR_Request*: 0mA
                b. For subsequent Request: As defined in Step 4.f.i
            2. B9…0 (Maximum Operating Current) = Maximum Current in the Source PDO.
        vii. For EPR AVS APDO:
            1. B20…9 (Output Voltage) = Maximum Voltage offered in Source APDO
            2. B6…0 (Operating Current):
                a. For the first *EPR_Request*: 0mA

b. For subsequent Request: As defined in Step 4.e.i

The initial *EPR_Request* is for PDO #8 (28V Fixed) at 0mA. The complete list of *EPR_Request* is defined in step 4.e.

b. The check fails if either *Accept, Reject* or *Wait* Message is not received, and the test stops here. If the *Reject* Message is received, the Tester reports the check with a warning, and the test stops here. [TEST.PD.PS.EPR.SRC3.1#11]

c. If the UUT responds with *Wait* Message, the Tester resends *Request* Message every *tSinkRequest* for ***Error! Reference source not found.*** times or until *Accept* Message is received. The check fails if the UUT responds with *Wait* Message after ***Error! Reference source not found.*** retries, and the test stops here. [TEST.PD.PS.EPR.SRC3.1#13]

d. The check fails if the first bit of Preamble of the *PS_RDY* Message is not received within *tSrcTransReq* from the last bit of the *GoodCRC* Message EOP corresponding to a transmitted *Accept* Message. [TEST.PD.PS.EPR.SRC3.1#12]

e. The Tester performs the following checks:

   i. If the transition involves a current decrease (but no PDO change), the Tester decreases current to the new value within *tSnkNewPower* min of the last bit of the *GoodCRC* sent in response to the *Accept* Message. The Tester draws the requested Operating Current with a current transition slew rate of 100mA per µs. The Tester monitors V_{BUS} voltage and the check fails if V_{BUS} voltage is not within the limits of *vSrcNew* or *vAvsNew*. [TEST.PD.PS.EPR.SRC3.1#3]

   ii. If the transition involves a current increase (but no PDO change), the Tester increases current to the new value after the last bit of the *GoodCRC* sent in response to the *PS_RDY* Message. The Tester draws the requested Operating Current with a current transition slew rate of 100mA per µs. The Tester monitors V_{BUS} voltage and the check fails if V_{BUS} voltage is not within the limits of *vSrcNew* or *vAvsNew*. [TEST.PD.PS.EPR.SRC3.1#4]

   iii. If the transition involves a change of PDO with a voltage increase, decrease the current drawn by the Tester to less than *iSnkStdby* within *tSnkStdby* of the last bit of the *GoodCRC* sent in response to the *Accept* Message. If the transition involves a change of PDO with decreasing voltage and current, the Tester decreases its current draw to the requested current or power for Fixed, Battery, Variable and SPR_AVS PDOs, 80% of the requested current for PPS PDO within *tSrcTransition* min after the end of the *GoodCRC* sent in response to the *Accept* Message. The Tester performs the following checks:

      1. The Tester checks that V_{BUS} remains within *vSrcNew* or *vAvsNew* of the starting voltage within *tSrcTransition* min after the end of the *GoodCRC* sent in response to the *Accept* Message. [TEST.PD.PS.EPR.SRC3.1#5]
      2. The Tester checks that the V_{BUS} transition meets *vSrcSlewPos* and *vSrcSlewNeg*. [TEST.PD.PS.EPR.SRC3.1#6]. (See Error! Reference source not found.)
      3. During the voltage transition, the Tester checks that the V_{BUS} voltage remains within *vSrcValid* limits from the time of crossing into the *vSrcValid* limits until *tSrcReady* max. [TEST.PD.PS.EPR.SRC3.1#7]
      4. The Tester checks that the V_{BUS} voltage remains within *vSrcNew* or *vAvsNew* limits between *tSrcReady* max and 100ms after *tSrcReady* max. [TEST.PD.PS.EPR.SRC3.1#8]
      5. The Tester checks that the UUT does not send *PS_RDY* before V_{BUS} is within *vSrcNew* or *vAvsNew*. [TEST.PD.PS.EPR.SRC3.1#9]
      6. After having sent *GoodCRC* in response to *PS_RDY*, if the transition involves a current increase, the Tester increases its current draw to the current requested and The Tester checks that the V_{BUS} voltage remains within *vSrcNew* or *vAvsNew* limits for 100ms after receiving *PS_RDY* [TEST.PD.PS.EPR.SRC3.1#10]

f. The Tester repeats Step-4a to Step-4d, replacing the appropriate fields with the next value from the following list. The Tester sends 9 separate *EPR_Request* Messages for each EPR_PDO.

For Fixed PDOs, replace B19…10 (Operating Current) and For EPR AVS APDO, replace B6…0 (Operating Current):

1. 0mA [performed in Step 4.b; listed here for reference only]
2. 25% of Maximum Current
3. 50% of Maximum Current
4. 75% of Maximum Current
5. 100% of Maximum Current
6. 75% of Maximum Current
7. 50% of Maximum Current
8. 25% of Maximum Current

To Text:

Description: As a Sink only, the Tester verifies that the Source UUT responds correctly to various *EPR_Request* Messages.
Test Specific Tester Behavior:
Test Conditions:

This test is applicable if the VIF parameter EPR_Supported_*As_Src* is set to Yes.

	Consumer Only	Provider Only, P/C, C/P, DRP
Rev3ChkdSrc		✓

Test Requirements: If Captive_Cable = YES, *vSrcNew* min or *vAvsNew* min limit shall be (*vSrcNew* min) - (0.75*I/X) or (vAvsNew min) - (0.75*I/X), where I is the actual current being drawn, and X=3 if the sourced PDO contract is within the range (0, 3A], or X=5 if the sourced PDO contract is within the range (3A, 5A)

Test Procedures:

1. There are 2 possible bring-up procedures:
    a. The UUT has VIF field PD_Port_Type set to Consumer/Provider, the Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.7.
    b. The UUT has VIF field PD_Port_Type set to anything else, the Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.1.
2. The check fails if the "EPR mode capable" field is set to 0 in the *Source_Capabilities* Message sent by UUT during the bring up procedure and the test stops here. [TEST.PD.PS.EPR.SRC3.1#1]
3. The Tester sends an *EPR_Mode* (Enter) Message with the Action field set to 0x01(Enter) and Data field set to 140 (140W) and enters EPR mode using common procedure COMMON.PROC.PD3.5. [TEST.PD.PS.EPR.SRC3.1#2]
4. The Tester repeats the following steps for each of the EPR_PDOs only (in the EPR_*Source_Capabilities* Message), starting with the 8th PDO:
    a. The Tester sends an *EPR_Request* Message.
        i. B27 (Reserved) = 0b
        ii. B26 (Capability Mismatch) = 0b
        iii. B25 (USB Communication Capable) = 0b
        iv. B24 (No USB Suspend) = 1b
        v. B22 (EPR Mode Capable) = 1b
        vi. For Fixed PDO:
            1. B19…10 (Operating Current):
                a. For first *EPR_Request*: 0mA
                b. For subsequent Request: As defined in Step 4.f.i
            2. B9…0 (Maximum Operating Current) = Maximum Current in the Source PDO.
        vii. For EPR AVS APDO:

1. B20…9 (Output Voltage) = Maximum Voltage offered in Source APDO
2. B6…0 (Operating Current):
    a. For the first *EPR_Request*: 0mA
    b. For subsequent Request: As defined in Step 4.e.i

The initial *EPR_Request* is for PDO #8 (28V Fixed) at 0mA. The complete list of *EPR_Request* is defined in step 4.e.

b. The check fails if any message received except *Accept* or *Wait*. [TEST.PD.PS.EPR.SRC3.1#11]
c. If the UUT responds with *Wait* Message, the Tester resends *Request* Message every *tSinkRequest* for ***Error! Reference source not found.*** times or until *Accept* Message is received. The check fails if the UUT responds with *Wait* Message after ***Error! Reference source not found.*** retries, and the test stops here. [TEST.PD.PS.EPR.SRC3.1#13]
d. The check fails if the first bit of Preamble of the *PS_RDY* Message is not received within *tSrcTransReq* from the last bit of the *GoodCRC* Message EOP corresponding to a transmitted *Accept* Message. [TEST.PD.PS.EPR.SRC3.1#12]
e. The Tester performs the following checks:
    i. If the transition involves a current decrease (but no PDO change), the Tester decreases current to the new value within *tSnkNewPower* min of the last bit of the *GoodCRC* sent in response to the *Accept* Message. The Tester draws the requested Operating Current with a current transition slew rate of 100mA per µs. The Tester monitors V_{BUS} voltage and the check fails if V_{BUS} voltage is not within the limits of *vSrcNew* or *vAvsNew*. [TEST.PD.PS.EPR.SRC3.1#3]
    ii. If the transition involves a current increase (but no PDO change), the Tester increases current to the new value after the last bit of the *GoodCRC* sent in response to the *PS_RDY* Message. The Tester draws the requested Operating Current with a current transition slew rate of 100mA per µs. The Tester monitors V_{BUS} voltage and the check fails if V_{BUS} voltage is not within the limits of *vSrcNew* or *vAvsNew*. [TEST.PD.PS.EPR.SRC3.1#4]
    iii. If the transition involves a change of PDO with a voltage increase, decrease the current drawn by the Tester to less than *iSnkStdby* within *tSnkStdby* of the last bit of the *GoodCRC* sent in response to the *Accept* Message. If the transition involves a change of PDO with decreasing voltage and current, the Tester decreases its current draw to the requested current or power for Fixed, Battery, Variable and SPR_AVS PDOs, 80% of the requested current for PPS PDO within *tSrcTransition* min after the end of the *GoodCRC* sent in response to the *Accept* Message. The Tester performs the following checks:
        1. The Tester checks that V_{BUS} remains within *vSrcNew* or *vAvsNew* of the starting voltage within *tSrcTransition* min after the end of the *GoodCRC* sent in response to the *Accept* Message. [TEST.PD.PS.EPR.SRC3.1#5]
        2. The Tester checks that the V_{BUS} transition meets *vSrcSlewPos* and *vSrcSlewNeg*. [TEST.PD.PS.EPR.SRC3.1#6]. (See Error! Reference source not found.)
        3. During the voltage transition, the Tester checks that the V_{BUS} voltage remains within *vSrcValid* limits from the time of crossing into the *vSrcValid* limits until *tSrcReady* max. [TEST.PD.PS.EPR.SRC3.1#7]
        4. The Tester checks that the V_{BUS} voltage remains within *vSrcNew* or *vAvsNew* limits between *tSrcReady* max and 100ms after *tSrcReady* max. [TEST.PD.PS.EPR.SRC3.1#8]
        5. The Tester checks that the UUT does not send *PS_RDY* before V_{BUS} is within *vSrcNew* or *vAvsNew*. [TEST.PD.PS.EPR.SRC3.1#9]
        6. After having sent *GoodCRC* in response to *PS_RDY*, if the transition involves a current increase, the Tester increases its current draw to the current requested and The Tester checks that the V_{BUS} voltage remains within *vSrcNew* or *vAvsNew* limits for 100ms after receiving *PS_RDY* [TEST.PD.PS.EPR.SRC3.1#10]

f. The Tester repeats Step-4a to Step-4d, replacing the appropriate fields with the next value from the following list. The Tester sends 9 separate *EPR_Request* Messages for each EPR_PDO.
   i. For Fixed PDOs, replace B19…10 (Operating Current) and For EPR AVS APDO, replace B6…0 (Operating Current):
      1. 0mA [performed in Step 4.b; listed here for reference only]
      2. 25% of Maximum Current
      3. 50% of Maximum Current
      4. 75% of Maximum Current
      5. 100% of Maximum Current
      6. 75% of Maximum Current
      7. 50% of Maximum Current
      8. 25% of Maximum Current

(b). TEST.PD.PS.EPR.SRC3.2 PDO Transition in EPR Mode

From:

Description: As a Sink only, the Tester enumerates all possible transitions among the PDOs of the UUT and verifies the UUT performs PDO transitions correctly in EPR Mode.
Test Specific Tester Behavior: N/A
Test Conditions:

This test is not applicable if the VIF parameter EPR_Supported_*As_Src* is set to No

	Consumer Only	Provider Only, P/C, C/P, DRP
Rev3ChkdSrc		✓

Test Procedures:

1. There are 2 possible bring-up procedures:
   a. The UUT has VIF field PD_Port_Type set to Consumer/Provider, the Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.7.
   b. The UUT has VIF field PD_Port_Type set to anything else, the Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.1.
2. The check fails if the "EPR mode capable" field is set to 0 in the *Source_Capabilities* Message sent by UUT during the bring up procedure and the test stops here. [TEST.PD.PS.EPR.SRC3.2#1]
3. The Tester sends an *EPR_Mode* (Enter) Message with the Action field set to 0x01(Enter) and Data field set to 140 (140W) and enters EPR mode using common procedure COMMON.PROC.PD3.5[TEST.PD.PS.EPR.SRC3.2#2]
4. The Tester repeats the following steps using the PDO as provided in the EPR_PDO Sequence Table (see Error! Reference source not found.). *NOTE*: PDO Transitions only applicable for valid PDOs in the *EPR_Source_Capabilities*. Ignore the unused PDOs. For AVS APDO, request the max, min, and mid voltages in sequence.
   a. The Tester sends an *EPR_Request* Message:
      i. B27 (Reserved) = 0b
      ii. B26 (Capability Mismatch) = 0b
      iii. B25 (USB Communication Capable) = 0b
      iv. B24 (No USB Suspend) = 1b
      v. B22 (EPR Mode Capable) = 1b
      v. For Fixed PDO:
         B19…10 (Operating Current) = 100mA
         B9…0 (Maximum Operating Current) = 100mA

For EPR AVS PDO:
B20…9 (Output Voltage) = Maximum Voltage offered in Source APDO
B7…0 (Operating Current) = 100mA

b. The check fails if either *Accept*, *Reject* or *Wait* Message is not received, and the test stops here. If *Reject* Message is received, the Tester reports the check with a warning, and the test stops here. [TEST.PD.PS.EPR.SRC3.2#3]

c. If the UUT responds with *Wait* Message, the Tester resends *Request* Message every *tSinkRequest* for ***Error! Reference source not found.*** times or until *Accept* Message is received. The check fails if the UUT responds with *Wait* Message after ***Error! Reference source not found.*** retries. [TEST.PD.PS.EPR.SRC3.2#10]

d. The check fails if the first bit of Preamble of the *PS_RDY* Message is not received within *tSrcTransReq* from the last bit of the *GoodCRC* Message EOP corresponding to a transmitted *Accept* Message. [TEST.PD.PS.EPR.SRC3.2#4]

e. The Tester performs the following checks:
Decrease the current drawn by the Tester to less than *iSnkStdby* within *tSnkStdby* min of the last bit of the *GoodCRC* sent in response to the *Accept* Message. The Tester performs the following checks:

1. The Tester checks the V_{BUS} is within *vSrcNew* or *vAvsNew* of the starting voltage within *tSrcTransition* min after the end of the *GoodCRC* sent in response to the *Accept* Message. [TEST.PD.PS.EPR.SRC3.2#5]
2. The Tester checks that the V_{BUS} transition meets *vSrcSlewPos* and *vSrcSlewNeg*. [TEST.PD.PS.EPR.SRC3.2#6] (See Error! Reference source not found.)
3. During the voltage transition, the Tester checks that the V_{BUS} voltage remains within *vSrcValid* limits from the time of crossing into the *vSrcValid* limits until *tSrcReady* max. [TEST.PD.PS.EPR.SRC3.2#7]
4. The Tester checks that the V_{BUS} voltage is within *vSrcNew* or *vAvsNew* limits between *tSrcReady* max and 100ms after *tSrcReady* max. [TEST.PD.PS.EPR.SRC3.2#8]
5. The Tester checks that the UUT does not send *PS_RDY* before the V_{BUS} is within *vSrcNew* or *vAvsNew*. [TEST.PD.PS.EPR.SRC3.2#9]

To:

Description: As a Sink only, the Tester enumerates all possible transitions among the PDOs of the UUT and verifies the UUT performs PDO transitions correctly in EPR Mode.
Test Specific Tester Behavior: N/A
Test Conditions:

This test is not applicable if the VIF parameter EPR_Supported_*As_Src* is set to No

	Consumer Only	Provider Only, P/C, C/P, DRP
Rev3ChkdSrc		✓

Test Procedures:

1. There are 2 possible bring-up procedures:
   a. The UUT has VIF field PD_Port_Type set to Consumer/Provider, the Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.7.

b. The UUT has VIF field PD_Port_Type set to anything else, the Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.1.

2. The check fails if the "EPR mode capable" field is set to 0 in the *Source_Capabilities* Message sent by UUT during the bring up procedure and the test stops here. [TEST.PD.PS.EPR.SRC3.2#1]
3. The Tester sends an *EPR_Mode* (Enter) Message with the Action field set to 0x01(Enter) and Data field set to 140 (140W) and enters EPR mode using common procedure COMMON.PROC.PD3.5[TEST.PD.PS.EPR.SRC3.2#2]
4. The Tester repeats the following steps using the PDO as provided in the EPR_PDO Sequence Table (see Error! Reference source not found.). *NOTE*: PDO Transitions only applicable for valid PDOs in the *EPR_Source_Capabilities*. Ignore the unused PDOs. For AVS APDO, request the max, min, and mid voltages in sequence.
   a. The Tester sends an *EPR_Request* Message:
      i. B27 (Reserved) = 0b
      ii. B26 (Capability Mismatch) = 0b
      iii. B25 (USB Communication Capable) = 0b
      iv. B24 (No USB Suspend) = 1b
      v. B22 (EPR Mode Capable) = 1b
      vi. For Fixed PDO:
          B19…10 (Operating Current) = 100mA
          B9…0 (Maximum Operating Current) = 100mA
          For EPR AVS PDO:
          B20…9 (Output Voltage) = Maximum Voltage offered in Source APDO
          B7…0 (Operating Current) = 100mA
   b. The check fails if any message received except *Accept* or *Wait*.[TEST.PD.PS.EPR.SRC3.2#3]
   c. If the UUT responds with *Wait* Message, the Tester resends *Request* Message every *tSinkRequest* for ***Error! Reference source not found.*** times or until *Accept* Message is received. The check fails if the UUT responds with *Wait* Message after ***Error! Reference source not found.*** retries. [TEST.PD.PS.EPR.SRC3.2#10]
   d. The check fails if the first bit of Preamble of the *PS_RDY* Message is not received within *tSrcTransReq* from the last bit of the *GoodCRC* Message EOP corresponding to a transmitted *Accept* Message. [TEST.PD.PS.EPR.SRC3.2#4]
   e. The Tester performs the following checks:
      Decrease the current drawn by the Tester to less than *iSnkStdby* within *tSnkStdby* min of the last bit of the *GoodCRC* sent in response to the *Accept* Message. The Tester performs the following checks:
      1. The Tester checks the V_{BUS} is within *vSrcNew* or *vAvsNew* of the starting voltage within *tSrcTransition* min after the end of the *GoodCRC* sent in response to the *Accept* Message. [TEST.PD.PS.EPR.SRC3.2#5]
      2. The Tester checks that the V_{BUS} transition meets *vSrcSlewPos* and *vSrcSlewNeg*. [TEST.PD.PS.EPR.SRC3.2#6] (See Error! Reference source not found.)
      3. During the voltage transition, the Tester checks that the V_{BUS} voltage remains within *vSrcValid* limits from the time of crossing into the *vSrcValid* limits until *tSrcReady* max. [TEST.PD.PS.EPR.SRC3.2#7]
      4. The Tester checks that the V_{BUS} voltage is within *vSrcNew* or *vAvsNew* limits between *tSrcReady* max and 100ms after *tSrcReady* max. [TEST.PD.PS.EPR.SRC3.2#8]
      5. The Tester checks that the UUT does not send *PS_RDY* before the V_{BUS} is within *vSrcNew* or *vAvsNew*. [TEST.PD.PS.EPR.SRC3.2#9]

(c). TEST.PD.EPR.SNK3.13 EPR PDO Transitions As Sink

USB PD CTS ENGINEERING CHANGE NOTICE

FROM:

Description: The Tester verifies that an EPR-capable Sink UUT, after entering EPR Mode, correctly responds to new EPR_Source_Capabilities Messages, requests valid EPR PDO/APDOs per its VIF, and that VBUS transitions stay within the EPR voltage envelopes.

Test Specific Tester Behavior:
Test Conditions:
This test is not applicable if the VIF parameter *EPR_Supported_As_Snk* is set to No

	Consumer Only	Provider Only	DRP, C/P, P/C
Rev3ChkdSnk	✓		✓
Rev3UnchkdSnk	✓		✓

Test Procedures:

1. There are 2 possible bring-up procedures:
   a. The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.8
   b. The UUT has VIF field PD_Port_Type set to Consumer Only, Consumer /Provider or DRP, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2
2. The Tester sends a *Get_Sink_Cap_Extended* Message to the UUT, the check fails if the UUT does not send a *Sink_Capabilities_Extended* Message. [TEST.PD.EPR.SNK3.13#1]
3. The Tester waits up to ***Error! Reference source not found.*** for the UUT to send an *EPR_Mode* (Enter, Action = 0x01) Message. If the VIF field EPR_Supported_As_Snk is Yes, the test stops here with a warning if no *EPR_Mode* Enter Message is received. [TEST.PD.EPR.SNK3.13#2]
4. If VIF field Captive_Cable = Yes, the Tester sends SOP' *Discover Identity* Command and checks the below fields in the response message, [TEST.PD.EPR.SNK3.13#3]
   a. V_{BUS} Current Handling Capability set to 10b(5A)
   b. Maximum V_{BUS} Voltage set to 11b(50V)
   c. EPR Mode Capable field set to 1b (EPR Mode Capable)
5. The Tester sends the *EPR_Mode* Message with Action field set to 0x03(Enter_Succeeded) and the Data field set to 0x0 within *tEnterEPR* from the *EPR_Mode* (Enter) Message.
6. The Tester sends the *EPR_Source_Capabilities* Message after 50ms from the last bit of the *GoodCRC* Message acknowledging the *EPR_Mode* (Enter_Succeeded) Message and establishes EPR contract using common procedure COMMON.PROC.PD3.3 from Step2. The Source_Capabilities Extended Message (sent on Get_Source_Cap_Extended) reports SPR Source PDP Rating = 100W and EPR Source PDP Rating = 240W. [TEST.PD.EPR.SNK3.13#4]
7. The Tester checks that the requested PDO/APDO is within the range of VIF Sink Capabilities Extended PDP for EPR mode: [TEST.PD.EPR.SNK3.13#5]
   a. For an EPR Fixed *EPR_Request* Message, the check fails if B19…10(Operating Current)*Voltage of the referenced PDO is greater than VIF field Snk_EPR_Maximum_PDP.

b. For an EPR_AVS *EPR_Request* Message, the check fails if:
   i. Snk_AVS_Supported = No and EPR_AVS *EPR_Request* is sent
   ii. Output Voltage is outside [15V, max offered EPR Fixed voltage], or
   iii. B20…9 (Output Voltage)*B6…0(Operating Current) is greater than VIF field Snk_EPR_Maximum_PDP.

8. The Tester intentionally sends a *Wait* Message. The test fails if another *EPR_Request* Message is received before *tSinkRequest* min. This is measured when the EOP of the *GoodCRC* acknowledging the *Wait* Message. [TEST.PD.EPR.SNK3.13#6]

9. If the UUT does not resend the *EPR_Request* Message within *tCtsInfinite*, the Tester resends the *EPR_Source_Capabilities* Message per step 6. The test fails if the UUT does not respond with an *EPR_Request* Message to the resent *EPR_Source_Capabilities*. [TEST.PD.EPR.SNK3.13#7]

10. The Tester sends an *Accept* Message to the *EPR_Request* Message received.

11. If the *EPR_Request* asks for an increasing-voltage transition:

    a. If the transition is to an EPR Fixed Supply PDO, or to the EPR AVS APDO with a voltage step greater than *vSmallStep*:

       i. Within *tSnkStdby* from the EOP of *GoodCRC* acknowledging the *Accept* Message, check that current drawn by the UUT does not exceed *iSnkStdby* max. [TEST.PD.EPR.SNK3.13#8]

       ii. Wait a nominal *tSrcTransition* from the *GoodCRC* and then change V_{BUS} voltage to the new requested value. For EPR Fixed and EPR AVS large-step transitions, the voltage change shall complete before *tSrcTransReq* expires.

       iii. As soon as the volage on V_{BUS} has stopped changing, check that current drawn by the UUT does not exceed *iSnkStdby* max until *PS_RDY* is sent. [TEST.PD.EPR.SNK3.13#9]

    b. If the transition is to EPR AVS with a voltage step less than or equal to *vSmallStep*:

       i. The Tester shall immediately begin changing V_{BUS} voltage after the *GoodCRC* Message acknowledging the *Accept* Message. The tester shall complete the voltage transition and send *PS_RDY* before *tAvsSrcTransSmall* expires.

       ii. If the newly negotiated current is less than the previously contracted current, after *tSnkNewPower* from the EOP of *GoodCRC* acknowledging the *Accept* Message, check that the current drawn by the UUT does not exceed the newly contracted current (I_{NEW}). [TEST.PD.EPR.SNK3.13#10]

       iii. If the newly negotiated current is unchanged or increased, check that the current drawn by the UUT does not exceed the previously contracted current during the transition after the EOP of *GoodCRC* acknowledging the *Accept* Message until the *PS_RDY* is sent by the Tester. [TEST.PD.EPR.SNK3.13#11]

11. If the *EPR_Request* asks for an unchanged or decreasing voltage, or no new PDO is requested:
    a. If the newly negotiated current is unchanged or increased: after *tSrcTransReq* EPR Mode max time period (before *PS_RDY* is sent) from the *GoodCRC* acknowledging the *Accept* Message, check that the current drawn from V_{BUS} does not exceed the previously contracted current. [TEST.PD.EPR.SNK3.13#12]
    b. If the newly negotiated current is decreased: after *tSnkNewPower* from the EOP of GoodCRC acknowledging the Accept Message, check that the current drawn by the UUT does not exceed I_{NEW}. [TEST.PD.EPR.SNK3.13#13]

12. The Tester sends a *PS_RDY* Message to the UUT.

13. The Tester checks that the current drawn by the UUT does not exceed the level specified in the contracted PDO over the next **Error! Reference source not found.**. [TEST.PD.EPR.SNK3.13#14]

14. The Tester offers an *EPR_Source_Capabilities* Message with EPR PDOs removed from the initially offered *EPR_Source_Capabilities* PDOs in Step 6 with a reduced Maximum Voltage of 20V and Port Maximum PDP of 100W. The check fails if the UUT does not send an *EPR_Request* Message for an SPR-only PDO. If *EPR_Request* voltage is not same as the current contract, repeat steps 7-10. If the requested voltage is same as the current contract, test ends here. [TEST.PD.EPR.SNK3.13#15]

15. The Tester re-offers an *EPR_Source_Capabilities Message* with the Maximum Voltage of 48V as initially offered in Step 6. If *EPR_Request* voltage is not same as the current contract, repeat steps 7-10. If the requested voltage is same as the current contract, test ends here. [TEST.PD.EPR.SNK3.13#16]

TO:

Description: The Tester verifies that an EPR-capable Sink UUT, after entering EPR Mode, correctly responds to new EPR_Source_Capabilities Messages, requests valid EPR PDO/APDOs per its VIF, and that VBUS transitions stay within the EPR voltage envelopes.

Test Specific Tester Behavior:
Test Conditions:
This test is not applicable if the VIF parameter *EPR_Supported_As_Snk* is set to No

	Consumer Only	Provider Only	DRP, C/P, P/C
Rev3ChkdSnk	✓		✓
Rev3UnchkdSnk	✓		✓

Test Procedures:

1. There are 2 possible bring-up procedures:

   c. The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.8

   d. The UUT has VIF field PD_Port_Type set to Consumer Only, Consumer/Provider or DRP, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2

2. The Tester checks the "EPR mode capable " field is set to 1 in the *Request* Message sent by the UUT during the bring-up procedure. If this field is not set to 1, the Tester waits for Time_To_EPR_Ready maximum for another *Request* Message. The test fails if no *Request* is received with "EPR mode capable" field is set, and the test stops here. [TEST.PD.EPR.SNK3.13#17]

3. The Tester sends a *Get_Sink_Cap_Extended* Message to the UUT, the check fails if the UUT does not send a *Sink_Capabilities_Extended* Message. [TEST.PD.EPR.SNK3.13#1]

4. The Tester waits for *EPR_Mode* (Enter, Action = 0x01) Enter Message for tCtsInfinite time. If the *EPR_Mode* Message is not received during this time, the check fails and the test stops here [TEST.PD.EPR.SNK3.13#2]

5. If VIF field Captive_Cable = Yes, the Tester sends SOP' *Discover Identity* Command and checks the below fields in the response message, [TEST.PD.EPR.SNK3.13#3]
    d. V_{BUS} Current Handling Capability set to 10b(5A)
    e. Maximum V_{BUS} Voltage set to 11b(50V)
    f. EPR Mode Capable field set to 1b (EPR Mode Capable)

6. The Tester sends the *EPR_Mode* Message with Action field set to 0x03(Enter_Succeeded) and the Data field set to 0x0 within *tEnterEPR* from the *EPR_Mode* (Enter) Message.

7. The Tester sends the *EPR_Source_Capabilities* Message after 50ms from the last bit of the *GoodCRC* Message acknowledging the *EPR_Mode* (Enter_Succeeded) Message and establishes EPR contract using common procedure COMMON.PROC.PD3.3 from Step2. The Source_Capabilities Extended Message (sent on Get_Source_Cap_Extended) reports SPR Source PDP Rating = 100W and EPR Source PDP Rating = 240W. [TEST.PD.EPR.SNK3.13#4]

8. The Tester checks that the requested PDO/APDO is within the range of VIF Sink Capabilities Extended PDP for EPR mode: [TEST.PD.EPR.SNK3.13#5]

    a. For an EPR Fixed *EPR_Request* Message, the check fails if B19…10(Operating Current)*Voltage of the referenced PDO is greater than VIF field Snk_EPR_Maximum_PDP.

    b. For an EPR_AVS *EPR_Request* Message, the check fails if:
        i. Snk_AVS_Supported = No and EPR_AVS *EPR_Request* is sent
        ii. Output Voltage is outside [15V, max offered EPR Fixed voltage], or
        iii. B20…9 (Output Voltage)*B6…0(Operating Current) is greater than VIF field Snk_EPR_Maximum_PDP.

9. The Tester intentionally sends a *Wait* Message. The test fails if another *EPR_Request* Message is received before *tSinkRequest* min. This is measured when the EOP of the *GoodCRC* acknowledging the *Wait* Message. [TEST.PD.EPR.SNK3.13#6]

10. If the UUT does not resend the *EPR_Request* Message within *tCtsInfinite*, the Tester resends the *EPR_Source_Capabilities* Message per step 6. The test fails if the UUT does not respond with an *EPR_Request* Message to the resent *EPR_Source_Capabilities*. [TEST.PD.EPR.SNK3.13#7]

11. The Tester sends an *Accept* Message to the *EPR_Request* Message received.

12. If the *EPR_Request* asks for an increasing-voltage transition:

    a. If the transition is to an EPR Fixed Supply PDO, or to the EPR AVS APDO with a voltage step greater than *vSmallStep*:

        1. Within *tSnkStdby* from the EOP of *GoodCRC* acknowledging the *Accept* Message, check that current drawn by the UUT does not exceed *iSnkStdby* max. [TEST.PD.EPR.SNK3.13#8]

        2. Wait a nominal *tSrcTransition* from the *GoodCRC* and then change V_{BUS} voltage to the new requested value. For EPR Fixed and EPR AVS large-step transitions, the voltage change shall complete before *tSrcTransReq* expires.

3. As soon as the volage on V_{BUS} has stopped changing, check that current drawn by the UUT does not exceed *iSnkStdby* max until *PS_RDY* is sent. [TEST.PD.EPR.SNK3.13#9]

b. If the transition is to EPR AVS with a voltage step less than or equal to *vSmallStep*:

1. The Tester shall immediately begin changing V_{BUS} voltage after the *GoodCRC* Message acknowledging the *Accept* Message. The tester shall complete the voltage transition and send *PS_RDY* before *tAvsSrcTransSmall* expires.

2. If the newly negotiated current is less than the previously contracted current, after *tSnkNewPower* from the EOP of *GoodCRC* acknowledging the *Accept* Message, check that the current drawn by the UUT does not exceed the newly contracted current (I_{NEW}). [TEST.PD.EPR.SNK3.13#10]

3. If the newly negotiated current is unchanged or increased, check that the current drawn by the UUT does not exceed the previously contracted current during the transition after the EOP of *GoodCRC* acknowledging the *Accept* Message until the *PS_RDY* is sent by the Tester. [TEST.PD.EPR.SNK3.13#11]

12. If the *EPR_Request* asks for an unchanged or decreasing voltage, or no new PDO is requested:
    a. If the newly negotiated current is unchanged or increased: after *tSrcTransReq* EPR Mode max time period (before *PS_RDY* is sent) from the *GoodCRC* acknowledging the *Accept* Message, check that the current drawn from V_{BUS} does not exceed the previously contracted current. [TEST.PD.EPR.SNK3.13#12]
    b. If the newly negotiated current is decreased: after *tSnkNewPower* from the EOP of GoodCRC acknowledging the Accept Message, check that the current drawn by the UUT does not exceed I_{NEW}. [TEST.PD.EPR.SNK3.13#13]

13. The Tester sends a *PS_RDY* Message to the UUT.

14. The Tester checks that the current drawn by the UUT does not exceed the level specified in the contracted PDO over the next **Error! Reference source not found.**. [TEST.PD.EPR.SNK3.13#14]

16. The Tester offers an *EPR_Source_Capabilities* Message with EPR PDOs removed from the initially offered *EPR_Source_Capabilities* PDOs in Step 6 with a reduced Maximum Voltage of 20V and Port Maximum PDP of 100W. The check fails if the UUT does not send an *EPR_Request* Message for an SPR-only PDO. If *EPR_Request* voltage is not same as the current contract, repeat steps 7-10. If the requested voltage is same as the current contract, test ends here. [TEST.PD.EPR.SNK3.13#15]

17. The Tester re-offers an *EPR_Source_Capabilities Message* with the Maximum Voltage of 48V as initially offered in Step 6. If *EPR_Request* voltage is not same as the current contract, repeat steps 7-10. If the requested voltage is same as the current contract, test ends here. [TEST.PD.EPR.SNK3.13#16]